

KAZACHINSK, Russian Federation

WMO: 303370

Lat: **56.280N**Lon: **107.568E**Elev: **357**StdP: **97.11**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.9	-41.3	-45.7	0.0	-43.1	-43.4	0.1	-40.8	7.0	-16.0	6.1	-16.3	0.1	210	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	29.2	17.1	27.2	16.7	25.3	16.1	19.4	24.6	18.4	24.2	17.4	23.1	2.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	13.4	21.1	16.5	12.3	20.3	15.4	11.4	19.3	57.2	24.1	53.9	24.4	50.7	23.3	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
6.3	5.4	4.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-46.0	32.8	2.7	1.3	-47.9	33.7	-49.5	34.4	-51.0	35.2	-52.9	36.1		
			-45.3	21.2	2.0	1.3	-46.7	22.1	-47.9	22.9	-49.0	23.6	-50.5	24.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.7	-25.6	-20.3	-10.5	-0.1	7.3	15.1	17.9	15.2	7.3	-1.5	-14.9	-23.5
	DBStd	16.41	8.95	7.87	7.30	4.89	3.90	3.91	2.77	3.12	4.14	5.29	9.18	9.49
	HDD10.0	5244	1103	847	635	304	103	5	0	2	103	357	746	1038
	HDD18.3	7726	1361	1081	893	554	342	110	42	104	331	616	996	1296
	CDD10.0	611	0	0	0	1	20	158	246	164	22	0	0	0
	CDD18.3	50	0	0	0	0	1	13	29	7	0	0	0	0
	CDH23.3	867	0	0	0	1	20	318	384	134	9	0	0	0
	CDH26.7	222	0	0	0	0	4	88	105	25	0	0	0	0
	WSAvg	1.6	1.1	1.2	1.7	2.1	2.3	1.7	1.5	1.5	1.6	1.8	1.6	1.2
	PrecAvg	412	20	14	10	15	33	55	70	70	48	28	26	23
(15) Precipitation	PrecMax	646	42	31	32	43	71	95	196	141	87	52	52	43
	PrecMin	319	6	5	0	2	8	17	13	14	2	1	8	8
	PrecStd	72	8	7	7	9	15	23	44	29	18	13	10	9
	PrecStd	72	8	7	7	9	15	23	44	29	18	13	10	9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	-3.1	0.6	9.6	17.8	27.1	31.6	32.2	29.2	24.7	14.0	4.6	-1.4	-1.4
	0.4% MCWB	-4.2	-2.3	2.8	7.7	14.0	17.4	18.3	17.9	15.0	6.9	1.7	-2.9	-2.9
	2% DB	-6.9	-2.3	5.8	13.9	22.7	29.2	29.6	26.7	20.3	10.5	1.8	-6.0	-6.0
	2% MCWB	-7.8	-4.2	0.7	5.6	11.2	16.4	18.1	17.1	12.7	5.1	-0.3	-6.9	-6.9
	5% DB	-10.3	-5.6	3.0	10.7	19.5	27.2	27.2	24.5	16.8	7.6	-0.1	-7.8	-7.8
	5% MCWB	-11.1	-6.7	-0.9	4.3	9.7	16.0	17.4	16.1	10.9	4.0	-1.6	-8.5	-8.5
	10% DB	-13.0	-8.4	0.7	8.1	16.5	24.8	25.3	22.2	14.1	5.2	-2.0	-10.1	-10.1
	10% MCWB	-13.7	-9.4	-2.2	2.8	8.2	15.0	17.0	15.5	9.7	2.4	-3.4	-10.6	-10.6
	0.4% WB	-4.1	-2.2	3.2	8.4	14.5	19.6	21.0	19.8	15.7	8.0	1.9	-2.8	-2.8
	0.4% MCDB	-3.1	0.7	9.1	15.5	25.3	25.1	26.3	24.8	22.5	12.8	3.3	-1.2	-1.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-7.8	-4.3	1.1	6.2	12.4	18.3	19.9	18.4	13.7	5.9	-0.1	-6.8	-6.8
	2% MCDB	-6.9	-2.6	5.1	12.6	20.2	25.9	24.6	24.0	18.6	9.3	1.4	-6.1	-6.1
	5% WB	-11.0	-6.7	-0.6	4.7	10.5	17.2	19.1	17.3	12.0	4.1	-1.5	-8.5	-8.5
	5% MCDB	-10.4	-5.6	2.7	10.0	17.7	24.1	24.0	22.2	15.6	7.2	0.1	-7.8	-7.8
	10% WB	-13.6	-9.4	-2.2	3.2	9.0	16.0	18.1	16.2	10.4	2.7	-3.2	-10.6	-10.6
	10% MCDB	-13.0	-8.4	0.6	7.8	15.6	23.2	23.7	20.9	13.5	5.0	-2.0	-10.1	-10.1
(35) Mean Daily Temperature Range	MDBR	9.8	13.8	16.1	12.6	12.8	13.8	12.1	11.7	10.3	8.7	9.4	9.3	9.3
	5% DB	9.3	13.5	16.0	16.5	19.4	19.7	17.6	17.0	15.9	13.4	8.7	9.1	9.1
	MCDBR	8.7	11.7	11.5	9.1	9.1	7.8	7.0	7.5	8.3	8.2	7.1	8.6	8.6
	5% WB	9.2	13.0	15.4	14.9	17.0	15.7	12.9	13.6	14.0	11.9	8.2	8.9	8.9
	MCWBR	8.6	11.4	11.2	8.7	8.6	7.0	6.3	6.8	7.7	7.6	6.8	8.3	8.3
	MCWBR	8.6	11.4	11.2	8.7	8.6	7.0	6.3	6.8	7.7	7.6	6.8	8.3	8.3
(40) Clear-Sky Solar Irradiance	taub	0.269	0.301	0.324	0.357	0.402	0.445	0.465	0.406	0.343	0.316	0.281	0.259	0.259
	taud	2.022	2.016	2.051	2.160	2.211	2.172	2.142	2.299	2.440	2.362	2.098	1.987	1.987
	Ebn at Noon	595	729	821	854	835	799	772	800	813	734	577	492	492
	Edh at Noon	64	97	121	128	130	137	139	111	84	69	59	51	51
(44) All-Sky Solar Radiation	RadAvg	0.82	1.79	3.25	4.64	5.16	5.65	5.20	4.32	2.78	1.47	0.81	0.55	0.55
	RadStd	0.07	0.12	0.15	0.32	0.38	0.45	0.56	0.33	0.26	0.19	0.10	0.06	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page